Project Design Phase-II

Data Flow Diagram & User Stories

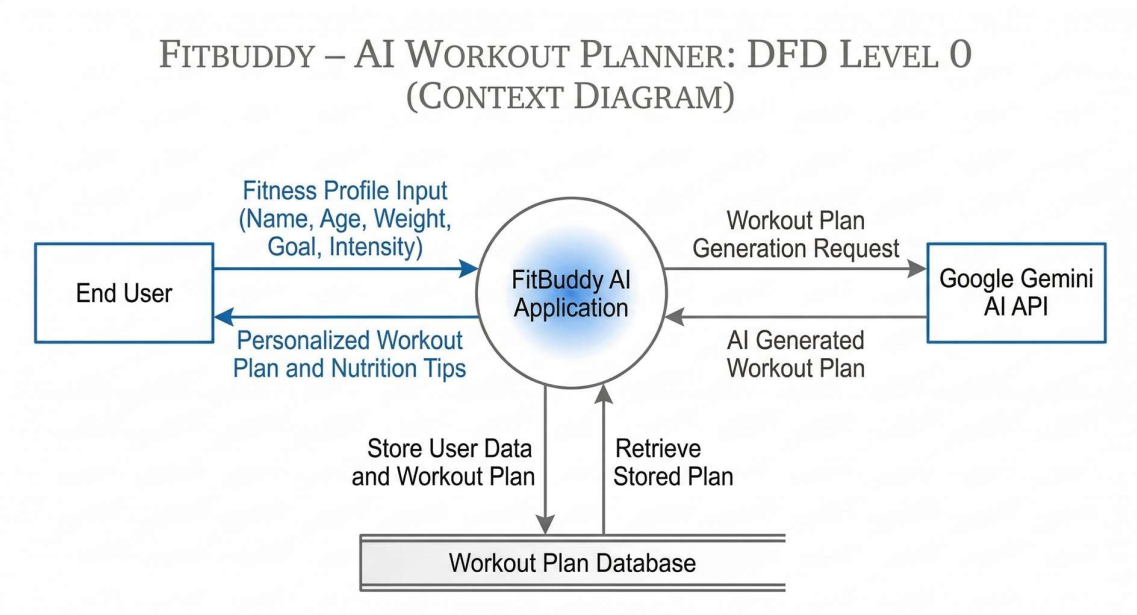
Date	31 January 2025
Team ID	GACSE-114
Project Name	FitBuddy – AI-Powered Fitness Planner
Maximum Marks	4 Marks

Data Flow Diagrams:

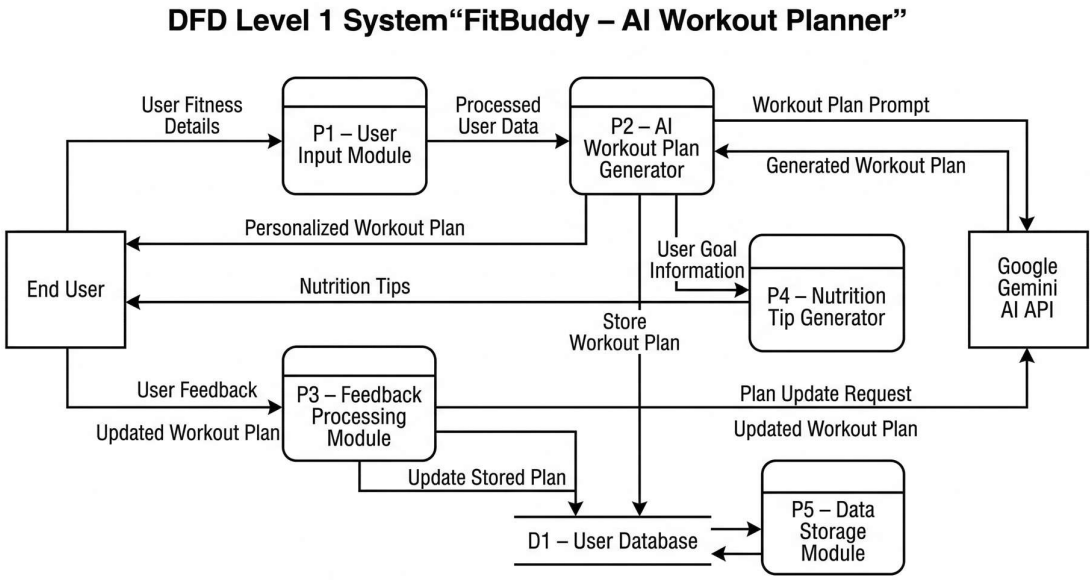
A Data Flow Diagram (DFD) represents how information flows through the FitBuddy system, how data enters, how it's processed, and where it is stored or shown to the user.

Data Flow Diagrams:

DFD Level 0:



DFD Level 1:



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
End User	User Profile Input	USN-1	As a user, I can enter my personal details (name, age, weight, height).	System should accept details and store them in session_state.	High	Sprint-1

		USN-2	As a user, I can select my fitness goal (Weight Loss, Muscle Gain, Endurance, Maintain Weight).	Selected goal must be reflected in AI plan generation.	High	Sprint-1
		USN-3	As a user, I can select my diet type (Vegetarian, Vegan, Non-Vegetarian).	Diet type must influence the diet plan in the AI output.	High	Sprint-1
		USN-4	As a user, I can select my activity level.	Activity level must update TDEE calculation.	High	Sprint-1
End User	Health Calculations	USN-5	As a user, I can view my BMI based on weight and height.	BMI should be accurate as per formula and displayed instantly.	High	Sprint-1
		USN-6	As a user, I can view my daily calorie requirement (TDEE).	TDEE must be correct using the Mifflin-St Jeor formula.	High	Sprint-1
End User	AI Fitness Plan Generation	USN-7	As a user, I receive a personalized workout plan.	AI-generated workout plan is displayed clearly.	High	Sprint-2

		USN-8	As a user, I get macro breakdown and calorie targets.	AI output should include calories & macro percentages.	High	Sprint-2
		USN-9	As a user, I receive a diet plan matching my diet type.	AI must obey diet type constraints.	High	Sprint-2
End User	Weight Prediction	USN-10	As a user, I can view a 30-day projected weight trend	A Streamlit line chart must appear with predicted	Medium	Sprint-3
				values.		
End User	AI Chat	USN-11	As a user, I can chat with the AI to ask fitness or diet questions.	Chat messages and responses should appear in real-time.	High	Sprint-3
		USN-12	As a user, my chat history should persist within the session.	Session_state must store conversation until page reload.	Medium	Sprint-3